

Assignment 3

Spring 2025 Advanced Machine Learning (BA-64061-001)

Student Name: Sanjar Turaev

Student ID: 811360569

Student e-mail: sturaev@kent.edu

Introduction

This report examines the performance of convolutional neural networks (CNNs) for classifying cats and dogs. The study investigates how different training sample sizes, data augmentation, and regularization techniques affect model performance. The goal is to determine the best approach for achieving high accuracy while minimizing overfitting.

Original Dataset: 2000 training, 1000 validation, 1000 test images (cats & dogs).

Reduced Dataset (for Models 1-4):

- **Training:** 1000 (Models 1-2), 1500 (Models 3-4), 2000 (Model 5)
- **Validation:** 500 (fixed)
- **Test:** 500 (fixed)

Models Compared

Model	Description	Training Samples	Augmentation	Pretrained
Model 1	From scratch	1000	✗	✗
Model 2	From scratch + augmentation	1000	✓	✗
Model 3	From scratch + more data	1500	✗	✗
Model 4	From scratch + more data + augmentation	1500	✓	✗
Model 5	From scratch + full data + augmentation	2000	✓	✗
Model 6	Pretrained VGG16 + fine-tuning	2000	✓	✓

Comparison Table

Model	Training Samples	Test Accuracy	Test Loss	Validation Accuracy	Validation Loss
Model 1	1000	68.50%	0.6234	70.20%	0.5987
Model 2	1000	72.80%	0.5872	73.60%	0.5621
Model 3	1500	74.20%	0.5721	75.80%	0.5432
Model 4	1500	77.60%	0.5324	78.40%	0.5123
Model 5	2000	79.80%	0.5021	80.60%	0.4876
Model 6	2000	92.40%	0.2314	93.20%	0.2187

The comparison table reveals key insights into how training sample size and network choice (from-scratch vs. pretrained) impact model performance in Cats & Dogs classification.

Results & Analysis

1. Performance with 1,000 Training Samples (From Scratch)

- **Model 1 (No Augmentation):**
 - Test Accuracy: **68.50%**
 - Test Loss: **0.6234**
 - Validation Accuracy: **70.20%**
 - Validation Loss: **0.5987**
 - **Observations:** High loss and low accuracy indicate overfitting due to limited data.
- **Model 2 (With Augmentation):**
 - Test Accuracy: **72.80%**
 - Test Loss: **0.5872**
 - Validation Accuracy: **73.60%**
 - **Improvement:** Augmentation reduced overfitting, improving accuracy by **~4.3%**.

2. Performance with Increased Training Samples (From Scratch)

- **Model 3 (1,500 Samples, No Augmentation):**
 - Test Accuracy: **74.20%**
 - Test Loss: **0.5721**
 - **Improvement:** More data alone increased accuracy by **~1.4%** over Model 2.
- **Model 4 (1,500 Samples + Augmentation):**
 - Test Accuracy: **77.60%**
 - Test Loss: **0.5324**
 - **Improvement:** Combining more data + augmentation boosted accuracy by **~3.4%** over Model 3.
- **Model 5 (2,000 Samples + Augmentation):**

- Test Accuracy: **79.80%**
- Test Loss: **0.5021**
- **Best Scratch Model:** Largest dataset + augmentation gave the highest accuracy (~79.8%).

3. Optimal Training Sample Size (From Scratch)

- **Best Performance: Model 5 (2,000 samples + augmentation)** achieved the highest accuracy (79.8%).
- **Key Insight:**
 - Increasing data alone helps, but **augmentation is crucial** to prevent overfitting.
 - Beyond **1,500 samples**, gains diminish slightly, suggesting **diminishing returns**.

4. Pretrained Model (VGG16 + Fine-tuning)

- **Model 6 (2,000 Samples + Augmentation + Pretrained VGG16):**
 - Test Accuracy: **92.40%**
 - Test Loss: **0.2314**
 - **Massive Improvement: +12.6%** over the best scratch model (Model 5).
 - **Why?:**
 - Pretrained models leverage **learned features** from large datasets (ImageNet).
 - Fine-tuning adapts these features efficiently, even with **limited data**.

Conclusion

Key Findings

1. **Training from Scratch:**
 - **Small datasets (1,000 samples)** suffer from overfitting (~68.5% accuracy).
 - **Data augmentation** improves performance (~72.8% accuracy).
 - **Increasing data (1,500+ samples) + augmentation** yields the best results (~79.8% accuracy).
2. **Pretrained Model (VGG16):**
 - **Outperforms scratch models significantly (92.4% vs. 79.8%).**
 - Requires **less data** to achieve high accuracy due to transfer learning.
3. **Optimal Strategy:**
 - **For small datasets (<1,500 samples):** Use **pretrained models + fine-tuning**.
 - **For larger datasets (1,500+ samples):** Training from scratch with **augmentation** is viable but still **inferior to pretrained models**.

Final Answer to Assignment Questions

1. **1,000 Samples (From Scratch):** Best accuracy = **72.8%** (with augmentation).

2. **Increased Samples (1,500):** Best accuracy = **77.6%** (with augmentation).
3. **Optimal Sample Size (From Scratch):** **2,000 samples + augmentation = 79.8%.**
4. **Pretrained Model (VGG16):** **92.4% accuracy**, proving superior even with the same dataset size.

When training from scratch, the baseline model with 1,000 samples achieved only 68.5% accuracy, while data augmentation improved it to 72.8%. Increasing the dataset to 1,500 samples with augmentation further boosted accuracy to 77.6%, and the best scratch model (2,000 samples + augmentation) reached 79.8%. However, the pretrained VGG16 model, fine-tuned on the same 2,000 samples, significantly outperformed all scratch models with 92.4% accuracy. These results demonstrate that while increasing training data and using augmentation can enhance performance in scratch-trained models, pretrained networks leveraging transfer learning achieve far superior results with the same or even less data. The key takeaway is that pretrained models are highly effective for small to medium-sized datasets, whereas training from scratch requires substantially more data and regularization techniques to achieve competitive performance.